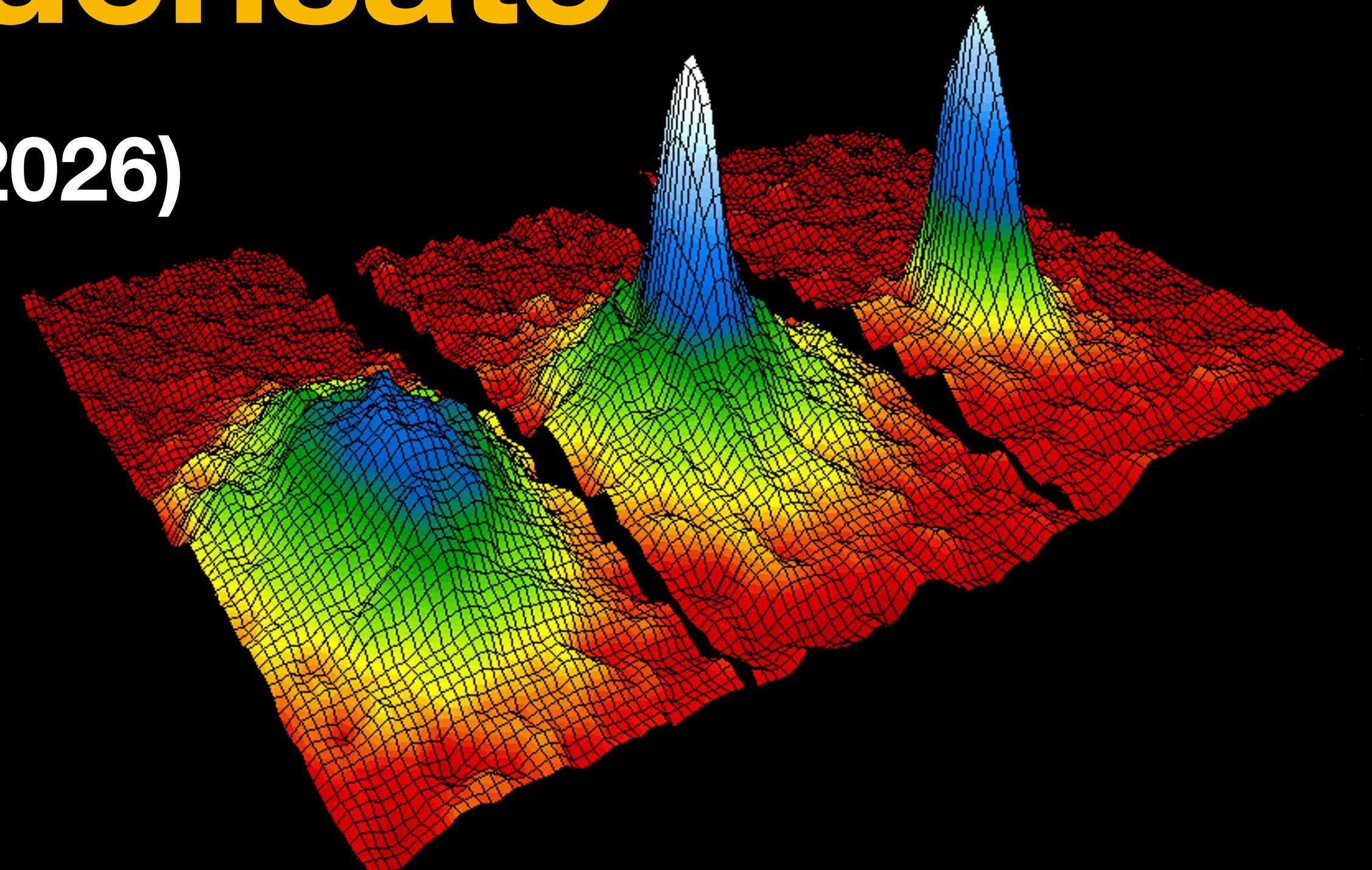


Introduction to Quantum Many-Body Physics with Bose-Einstein Condensate

UNIST Tutorial Session I (April 30th 2026)

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UNIST Tutorial Talk Prep.

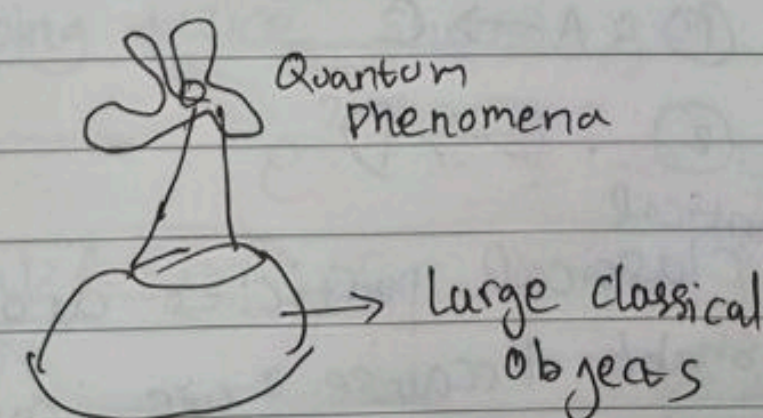
Tutorial #1

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Preface: Pushing the boundaries of QM

- QM started out as theory for microscopic system
 - Zoom into single particle, e.g. atom/subatomic particle
 - undergrad: particle in a box, particle in a harmonic potential, single electron in hydrogen atoms.
- We live in a world with extremely many particles.
 - More atoms in a handful of sand than there are stars in the Milky Way Galaxy.
 - If QM is to be able to explain the world we see, we must be able to do QM for many-particle system.
- Interestingly, when systems are large $\rightarrow$ Classical physics. How classical physics emerge out of QM is still a mystery. For example, measurement problem



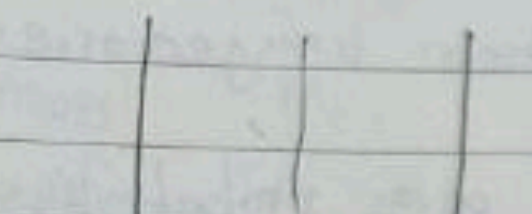
Quantum

Classical

- Huge effort in pushing the boundaries of QM into macro. system $\rightarrow$ BEC is at the frontier. The seed was planted in $\sim 1925 \rightarrow$ BEC was created in the lab ~ 1995 .

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where $S_{\pm}$ are projectors to symmetric and antisymmetric subspaces

- Two indistinguishable particles in a box example:



$$\phi_n(x) = \sqrt{\frac{2}{L}} \cos\left(\frac{n\pi x}{L}\right)$$

→ Two bosons occupying $n=1$ and $n=2$ state has a wavefunction

$$\begin{aligned} \Psi(x_1, x_2) &= \frac{1}{\sqrt{2}} (\phi_1(x_1)\phi_2(x_2) + \phi_1(x_2)\phi_2(x_1)) \\ &= \frac{\sqrt{2}}{L} \left(\cos(\pi x_1/L) \cos(2\pi x_2/L) + \cos(\pi x_2/L) \cos(2\pi x_1/L) \right) \end{aligned}$$

→ For two fermions

$$\Psi(x_1, x_2) = \frac{1}{\sqrt{2}} (\phi_1(x_1)\phi_2(x_2) - \phi_1(x_2)\phi_2(x_1))$$

→ Note that two fermions can't occupy the same state, e.g. $n=1$ for both gives

$$\Psi = 0 \Rightarrow \text{Pauli Exclusion Principle!}$$

No two fermions can occupy the same state

→ Note: Two fermions can occupy the same orbital if their spin are different

$$\mathcal{H} = \mathcal{H}_{\text{spatial}} \otimes \mathcal{H}_{\text{spin}}$$

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Basis states for $\mathcal{H}$ is given by

$$\{|\phi_n\rangle \otimes |s\rangle\} \text{ where } s \in \{\uparrow, \downarrow\}$$

and $\langle x|\phi_n\rangle = \phi_n(x)$. Consider a fermionic state

$$|\Psi\rangle = |\phi_n\rangle_1 \otimes |\phi_n\rangle_2 \otimes \frac{1}{\sqrt{2}} (|\uparrow\rangle_1 |\downarrow\rangle_2 - |\downarrow\rangle_1 |\uparrow\rangle_2)$$

Swapping $1 \leftrightarrow 2$ still gives $|\Psi\rangle \rightarrow -|\Psi\rangle$. The state

$\frac{1}{\sqrt{2}} (|\uparrow\rangle_1 |\downarrow\rangle_2 - |\downarrow\rangle_1 |\uparrow\rangle_2)$ is called singlet. For bosons,

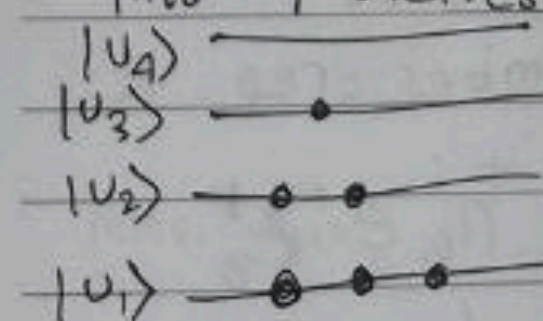
the spin part can be in any of the triplet states:

$$|\uparrow\uparrow\rangle, |\downarrow\downarrow\rangle, \frac{1}{\sqrt{2}} (|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle)$$

- One can generalize the logic for two particles here to the case of N particles using generalized symmetrized and anti-symmetrized wavefunction. But it is so tedious and rarely used in the formalism of many-body physics beyond few particles. There must be a simpler way.

Occupation Number Representation and Second Quantization

Consider the problem of distributing 6 identical balls into 4 shelves.

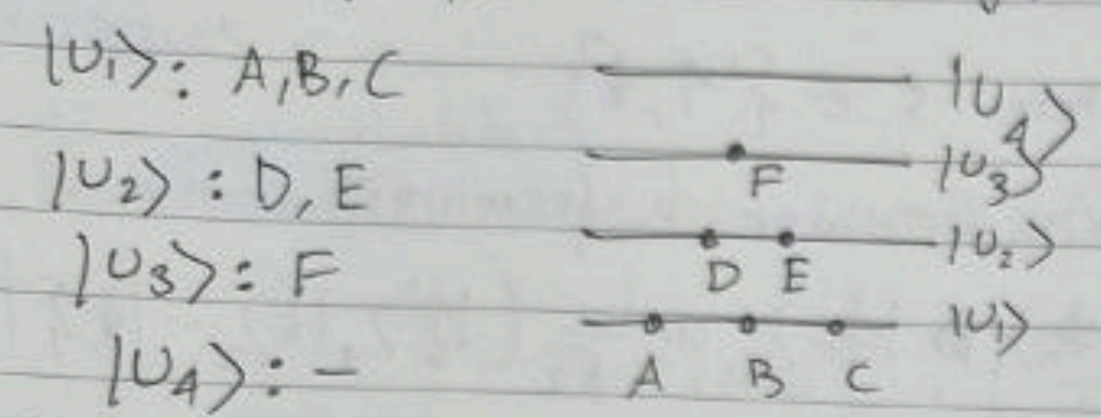


⇒ One possible state
Balls → Particles

Shelves → Single-particle

basis $\{ |u_i\rangle \} \in \mathcal{H}$

• How to specify this state? e.g.



What if I exchange $A \leftrightarrow D$?

$|u_1\rangle: D, B, C$
 $|u_2\rangle: A, E$
 $|u_3\rangle: F$
 $|u_4\rangle: -$

According to indistinguishability postulate, this state is identical to before for bosons, and carry minus sign for fermions

→ Redundancy of description! To describe indistinguishable quantum particles, we only need to track:

Bosons	Fermions
① # of particles on each $ u_i\rangle$	① # of particles on each $ u_i\rangle$
	② The permutation

→ Occupation number representation! (Second quantization)

Fix an ordering of basis $\{|u_i\rangle\} \in \mathcal{H}$ and the state in occupation number rep. is

$$|n_{u_1} n_{u_2} n_{u_3} \dots\rangle, n_{u_i} \in \mathbb{Z}_0^+$$

Fock state

• Example: $|3, 2, 1, 0\rangle \rightarrow$ This ~~was~~ state is only valid for bosons. Pauli exclusion principle forces fermionic Fock state to be

$$|n_{u_1} n_{u_2} \dots\rangle, n_{u_i} \in \{0, 1\}$$

• Q: What ~~vector~~ space do Fock states live in? Consider the following state:

$$|\psi\rangle = \underbrace{|000\rangle}_{0\text{-particle}} + \underbrace{|010\rangle}_{1\text{-particle}} + \underbrace{|020\rangle}_{2\text{-particle}}$$

$|\psi\rangle$ must reside in space with variable particle number!

Recall, N -particle state resides in $S_{\pm} \mathcal{H}^{\otimes N}$

A Fock state lives in:

$$\mathcal{F} = \mathcal{F}_0 \oplus \mathcal{F}_1 \oplus \dots = \bigoplus_{n=0}^{\infty} \mathcal{F}_n$$

where $\mathcal{F}_n = S_{\pm} \mathcal{H}^{\otimes n}$. $\mathcal{F}$ is called Fock space.

Bosonic Creation and Annihilation Operators

We will define creation and annihilation operators for bosons:

$$\begin{aligned} b_i^\dagger | \dots, n_i, \dots \rangle &= \sqrt{n_i + 1} | \dots, n_i + 1, \dots \rangle \\ b_i | \dots, n_i, \dots \rangle &= \sqrt{n_i} | \dots, n_i - 1, \dots \rangle \end{aligned}$$

The basis u is implicit to simplify notation

Note that $b_i^\dagger : \mathcal{F}_N \rightarrow \mathcal{F}_{N+1}$

$$b_i : \mathcal{F}_N \rightarrow \mathcal{F}_{N-1}$$

From these definitions, we can see that

$$b_i^\dagger b_i |\dots, n_i, \dots\rangle = n_i |\dots, n_i, \dots\rangle$$

↳ Number operator $N_i = b_i^\dagger b_i$ for state $|U\rangle_i$

• Commutation Relations (Boson)

$$b_i^\dagger b_j^\dagger |n_i, n_j\rangle = \sqrt{(n_i+1)(n_j+1)} |n_i+1, n_j+1\rangle \quad i \neq j$$

$$b_j^\dagger b_i^\dagger |n_i, n_j\rangle = \sqrt{(n_i+1)(n_j+1)} |n_i+1, n_j+1\rangle \quad -$$

$$[b_i^\dagger, b_j^\dagger] = 0 \quad i \neq j$$

↳ Easy to check that $[b_i^\dagger, b_i] = 0$

Thus, we have $[b_i^\dagger, b_j^\dagger] = [b_i, b_j] = 0$

$$b_i^\dagger b_j |n_i, n_j\rangle = \sqrt{(n_i+1)n_j} |n_i+1, n_j-1\rangle$$

$$b_j^\dagger b_i |n_i, n_j\rangle = \sqrt{(n_i+1)n_j} |n_i+1, n_j-1\rangle \quad -$$

$$[b_j, b_i^\dagger] = 0 \quad \text{for } i \neq j$$

$$b_i b_i^\dagger |n_i\rangle = (n_i+1) |n_i\rangle$$

$$b_i^\dagger b_i |n_i\rangle = n_i |n_i\rangle \quad -$$

$$[b_i, b_i^\dagger] = 1$$

Finally, we get bosonic commutation relations

$$\left[\begin{array}{l} [b_i, b_j] = [b_i^\dagger, b_j^\dagger] = 0 \\ [b_i, b_j^\dagger] = \delta_{ij} \end{array} \right] \Rightarrow \text{Enforce boson particle statistics}$$

- Any Fock state can be built by operating $\{b_i^\dagger\}$ on the vacuum state. $|\text{vac}\rangle = |0000\dots\rangle$

$$|n_1, n_2, \dots\rangle = \prod_{i=1} \frac{(b_i^\dagger)^{n_i}}{\sqrt{n_i!}} |\text{vac}\rangle$$

Note that $|\text{vac}\rangle$ is not "nothing", a vacuum can have non-zero expectation value $\langle \text{vac} | \hat{O} | \text{vac} \rangle \neq 0$ for some observables

Fermionic Creation and Annihilation Operators

- Remember that for fermions, a fock state is

$$|n_1, n_2, \dots\rangle \quad n_i \in \{0, 1\}$$

after fixing an ordering of single particle states $\{|u_i\rangle\} \in \mathcal{H}$. Changing the ordering cost a sign

$$\Rightarrow \text{Ex: } |11\rangle = |u_1 u_2\rangle = \frac{1}{\sqrt{2}} (|u_1\rangle |u_2\rangle - |u_2\rangle |u_1\rangle)$$

If instead I choose $|11\rangle = |u_2 u_1\rangle = -|u_1 u_2\rangle$

So changing ordering $\rightarrow$ minus sign

- Fermionic creation and annihilation operators are

$$c_i^\dagger \underbrace{|\dots, n_i, \dots\rangle}_{F_N} = (-1)^{S_i} (1 - n_i) \underbrace{|\dots, n_i+1, \dots\rangle}_{F_{N+1}}$$

$$c_i \underbrace{|\dots, n_i, \dots\rangle}_{F_N} = (-1)^{S_i} n_i \underbrace{|\dots, n_i-1, \dots\rangle}_{F_{N-1}}$$

where $S_i = \sum_{j < i} n_j$ is the Jordan-Wigner sign

that ensures consistency regardless of the chosen ordering.

- Note that by construction

$$\begin{aligned} c_i^\dagger | \dots, n_i, \dots \rangle &= 0 \text{ if } n_i = 1 \\ c_i | \dots, n_i, \dots \rangle &= 0 \text{ if } n_i = 0 \end{aligned} \quad \left. \vphantom{\begin{aligned} c_i^\dagger | \dots, n_i, \dots \rangle &= 0 \text{ if } n_i = 1 \\ c_i | \dots, n_i, \dots \rangle &= 0 \text{ if } n_i = 0 \end{aligned}} \right\} \text{Pauli exclusion}$$

- I will assert without proof that fermionic creation and annihilation operators satisfy anti-commutation relations

$$\{c_i^\dagger, c_j^\dagger\} = \{c_i, c_j\} = 0$$

$$\{c_i, c_j^\dagger\} = \delta_{ij}$$

where $\{A, B\} = AB + BA$ is the anti-commutator.

One-body operator

So far, we have only discussed how to write a many-body quantum state. To complete the formalism of QM in this new occupation number language we also need to know how to write operators. Let us start with the simplest case of "one-body operator", i.e. operator that acts on one particle at a time.

Consider first distinguishable particles:

$$\begin{aligned} \{ |u_q\rangle \} \in \mathcal{H} & \xrightarrow{\quad} \mathcal{H}_1 \otimes \mathcal{H}_2 \otimes \dots \otimes \mathcal{H}_N \\ \hat{f}_q & \quad \quad \quad \text{N-particles} \quad 1 \otimes 1 \otimes \dots \otimes \hat{f}_q \otimes \dots \otimes 1 \end{aligned}$$

But for indistinguishable particles, we cannot apply operator that acts on a specific particle (because then the particle would be distinguishable). Just like how we only admit symmetric/antisymmetric states, we also admit only symmetric/antisymmetric operators. The simplest of such operator is

$$\hat{F} = \sum_{q=1}^N \hat{f}_q \rightarrow \begin{array}{l} \text{one-body operator} \\ \text{acts on all N-particles} \\ \text{but only one particle at a time} \end{array}$$

Let's write down ~~the~~ $\hat{F}$ in first quantized form

$$\hat{f}_q = \sum_{k,l} f_{kl} |U_k\rangle_q \langle U_l| \quad f_{kl}^{(q)} = \langle U_k | \hat{f}_q | U_l \rangle = f_{kl}$$

$$\hat{F} = \sum_{q=1}^N \sum_{k,l} f_{kl} |U_k\rangle_q \langle U_l| \rightarrow \text{acts on state in } \mathcal{H}^{\otimes N}$$

I will again assert without proof that in second quantized form, this operator is

$$\hat{F} = \sum_{k,l} f_{kl} \underbrace{b_k^\dagger b_l}_{\text{for bosons}} \rightarrow \text{acts on state in } \mathcal{F}$$

$$\hat{F} = \sum_{k,l} f_{kl} \underbrace{c_k^\dagger c_l}_{\text{for fermions}}$$

Note that as usual, information about the basis states $\{ |u_q\rangle \}$ is hidden as $c_k = c_{u_k}$, $b_l = b_{u_l}$ and it is also encoded in $f_{kl} = \langle U_k | \hat{f} | U_l \rangle$.

• Examples of one-body operator:

① Particle number operator

$$\hat{N} = \sum_k b_k^\dagger b_k \quad (f_{kl} = \delta_{kl})$$

② Hamiltonian of non-interacting particles

Choose energy eigenbasis for a single particle

$$\hat{H} = \sum_{i=1}^N \hat{h}_i \quad (\text{first-quantized})$$

~~$$\hat{h}_i = \sum_k \epsilon_k |e_k\rangle \langle e_k|$$~~

$$\hat{h}_i = \sum_k \epsilon_k |e_k\rangle \langle e_k|$$

Then in second quantized form:

$$\hat{H} = \sum_k \epsilon_k b_k^\dagger b_k \quad (f_{kl} = \epsilon_k \delta_{kl})$$

↳ This Hamiltonian can describe a variety of non-interacting particles scenario.

→ Harmonic oscillators

→ Non-interacting hydrogen atoms

→ Free particles, non-interacting qubits

→ ... any cases where we know how to diagonalize the single particle problem.

~~...~~

Changing basis in second quantization

Start from two sets of single particle basis

$$\{|u_i\rangle\}, \langle u_i | u_k \rangle = \delta_{ik}$$

$$\{|v_i\rangle\}, \langle v_i | v_k \rangle = \delta_{ik}$$

$$b_{u_i}^\dagger |n_i\rangle = \sqrt{n_i+1} |n_i+1\rangle$$

$$b_{v_i}^\dagger |n_i\rangle = \sqrt{n_i+1} |n_i+1\rangle$$

$$b_{u_i} |n_i\rangle = \sqrt{n_i} |n_i-1\rangle$$

$$b_{v_i} |n_i\rangle = \sqrt{n_i} |n_i-1\rangle$$

similarly for fermions!

Changing basis $\longleftrightarrow$ Figuring out how $\{b_{u_i}\}$ and $\{b_{v_j}\}$ are related.

Consider $b_{u_i}^\dagger |0\rangle = |u_i\rangle$ $b_{v_j}^\dagger |0\rangle = |v_j\rangle$

→ Add resolution of identity in U basis

$$b_{v_j}^\dagger |0\rangle = \sum_i |u_i\rangle \langle u_i | v_j \rangle = \sum_i \langle u_i | v_j \rangle |u_i\rangle$$

$$= \sum_i \langle u_i | v_j \rangle b_{u_i}^\dagger |0\rangle$$

$$\therefore b_{v_j}^\dagger = \sum_i \langle u_i | v_j \rangle b_{u_i}^\dagger$$

Take adjoint on both sides

$$b_{v_j} = \sum_i \langle v_j | u_i \rangle b_{u_i}$$

Same logic holds for fermions

Homework/Exercise

Show that canonical commutation relation is preserved on any basis

$$[b_{u_i}, b_{v_j}^\dagger] = [b_{v_i}, b_{u_j}^\dagger]$$

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First to Second Quantization Summary

First Quant. $\Rightarrow$ state lives in ~~single particle~~ $\mathcal{H}$ (i.e. $\{|U_i\rangle\}$ Possible states can be discrete, N -identical particles live in $\mathcal{H}^{\otimes N}$)
 $\Rightarrow$ ~~N particles~~ indistinguishable particles must get symmetrized/antisymmetrized $S_{\pm} \mathcal{H}^{\otimes N}$

Second Quant. $\Rightarrow$ state is described in occupation number rep., lives in $\mathcal{F}$ (Fock space) $= \bigoplus_{\pm} \mathcal{H}^{\otimes n}$
 $\Rightarrow$ ~~Fundamental objects~~: states are constructed from $|vac\rangle$ with $b_i, b_i^\dagger$ (bosons) $c_i, c_i^\dagger$ with their respective commutation/anti commutation algebra.

Challenge: $\Rightarrow$ One-body operator, two-body operator, ...
 How to deal with interacting particles?

Hint: Two-body, three-body, ..., N -body operators.

See you next week!